On Stabilizing Communication Law for Bilateral Teleoperations

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***Abstract*—This paper puts forward a stabilizing communication law to enable direct environmental contact force feedback for the wave transformation-based teleoperation architecture, potentially improving the human operator's experience of telepresence. The stability of the established stabilizing communication law and the closed-loop teleoperation system will be discussed. The simulation results are presented to highlight the advantage of the proposed communication law over the lossless passive one.**

***Index Terms*—telerobotics, force reflecting, wave transformation, time-delay**

I. Introduction

A time-delay of tens of milliseconds might destabilize the otherwise stable bilateral teleoperation system with force feedback [1]. One of the main tools for overcoming the time-delay problem is passivity. The concept of passivity-based teleoperation was originated in [2] on the basis of scattering theory and later interpreted in the context of wave transformation (WT) [3]. Several improvements have been reported to address shortcomings in the aforementioned early works by mitigating time-varying delays (see [4], [5]), avoiding wave reflection (see [6], [7]), and correcting position drift (see [6], [8]). However, to keep the passivity of the remote subsystem, most existing WT-based teleoperation schemes must rely on coordinating force for feedback rather than real environmental contact force. Such an implementation drastically diminishes the teleoperation transparency. Often, the coordinating force is insensitive to the change of environmental contact force leading to inaccurate force information signaling back to the human operator. The inaccuracy is substantial for robots with large internal friction forces and/or high gear ratios. Direct contact force feedback, on the other hand, can significantly enhance the sense of telepresence. Nevertheless, the problem is direct contact force feedback causes contact instability even when the lossless WT is used [2]. It will be later demonstrated that, in such a circumstance, the remote subsystem is non-passive causing non-passivity to the overall closed-loop system, and hence the instability. To summarize, direct contact force feedback can improve the sense of telepresence for WT-based stabilization schemes, but the stability of the overall system is put at risk. Although some existing methods based on the small-gain theorem and wave filters can overcome this dilemma (see [9], [10]), they tend to be heuristic and their extensions to nonlinear teleoperations are not plausible.

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This paper introduces a strategy for enabling direct contact force feedback for the WT-based stabilization approach while maintaining the system's stability in the face of arbitrarily large constant time delays. We will design a new WT scheme with excessive levels of passivity to compensate for the non-passivity of the remote side. This is the fundamental idea of the paper. The traditional lossless WT, at its best, does not influence the overall system stability. Our proposed communication law based on the developed WT can indeed stabilize non-passive (remote) systems. The stability of a closed-loop system under the suggested WT formulation will be demonstrated using Lyapunov method.

To clarify, this paper extends the results presented in [11] on the following points

- Theorem 3 (section III), which demonstrates the stability of the established communication law, is added.
- Theorem 5 (section IV) is included to prove that the proposed WT-based bilateral teleoperation system can achieve the desired performance in free-motion and hard-contact scenarios.
- A key simulation (simulation 2, section V) is provided to highlight the advantage of the proposed stabilizing communication law over the conventional lossless passive WT scheme.

Notations: $\mathcal{L}_\infty$ and $\mathcal{L}_2$ denote the space of bounded functions and square integrable functions, respectively [12, p. 196]. The set of real numbers, non-negative real numbers, and non-negative integers are represented by $\mathbb{R}$, $\mathbb{R}^+$ and $\mathbb{N}$, respectively. The two-dimensional Euclidean space is $\mathbb{R}^2$. The time variable is t. Moreover, $\mathscr{C}$ denotes the space of all (possibly vector-valued) continuous functions defined on $\mathbb{R}$.

II. Problem formulation

In this section, we will show that the instability occurs when the contact force (rather than the coordinating force) is directly fed back in the WT-based teleoperation architecture of [2]. We demonstrate this using a simulation example and explain the reason. The teleoperation system considered throughout the paper is shown in Fig. 1. The system is identical to that of [2] but the contact force is transmitted back to the local side rather than the coordinating force. The dynamics of local robot is given by

$$M_m\ddot{x}_m + C_m\dot{x}_m = \tau_m + F_h, \tag{1}$$

where $M_m > 0$, $C_m \geq 0$, x_m, and τ_m denote the mass, the viscous damping coefficient, the position, and the applied motor force of the local robot, respectively. The human force

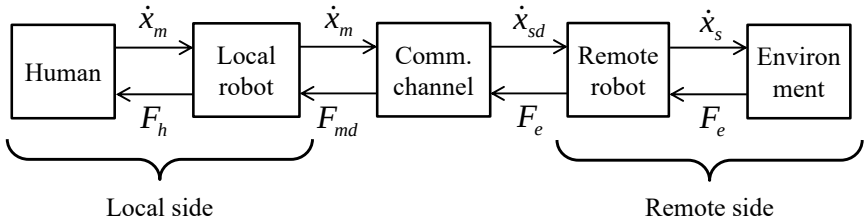

Fig. 1. A velocity–force teleoperation architecture with direct contact force feedback.

exerting on the local robot is represented by F_h, and it is modeled by

$$F_h = -K_h x_m + f_h^\star. \tag{2}$$

In (2), $K_h > 0$ is the spring coefficient of the human hand. The exogenous force $f_h^\star$ generated by human muscles can be decomposed into two components as follows

$$f_h^\star = K_h x_{md} + F_h^\star,$$

where $x_{md} \in \mathbb{R}$ and $F_h^\star$ stand for the constant and the time-varying part of $f_h^\star$, respectively. It is also assumed that $F_h^\star$ is bounded and has a finite energy, that is to say

$$F_h^\star \in \mathcal{L}_\infty \text{ and } \int_0^\infty F_h^{\star 2} dt < \infty. \tag{3}$$

The remote robot's dynamics is specified by

$$M_s \ddot{x}_s + C_s \dot{x}_s = \tau_s - F_e, \tag{4}$$

where $M_s > 0$, $C_s \geq 0$, x_s, and τ_s are the mass, the viscous damping coefficient, the position, and the applied motor force of the remote robot, respectively. It is supposed that the remote robot is in contact with a spring environment, having a coefficient of $K_e \geq 0$, i.e., the contact force F_e is

$$F_e = K_e x_s. \tag{5}$$

According to [2], the remote and local robots are controlled by the following laws

$$\tau_s = -B_{s1}\dot{x}_s + F_s, \quad B_{s1} > 0 \tag{6}$$

$$\tau_m = -B_m \dot{x}_m - F_{md}, \quad B_m > 0. \tag{7}$$

The coordinating force F_s in (6) is given by the below proportional–derivative (PD) controller

$$F_s = K_s \int_0^t (\dot{x}_{sd} - \dot{x}_s) d\tau + B_{s2} (\dot{x}_{sd} - \dot{x}_s), \tag{8}$$

with $K_s > 0$ and $B_{s2} > 0$. When a lossless WT is used for the communication channel, the reflecting force F_{md} in (7) and the velocity command $\dot{x}_{sd}$ in (8) are determined by the following communication law

$$\begin{aligned} \dot{x}_{sd}(t) &= \dot{x}_m(t-T) + 4b^2 \left[F_{md}(t-T) - F_e(t)\right] \\ F_{md}(t) &= F_e(t-T) + \frac{1}{4b^2}\left[\dot{x}_m(t) - \dot{x}_{sd}(t-T)\right], \end{aligned} \tag{9}$$

where $1/4b^2 > 0$ is the characteristic impedance and $T > 0$ is the magnitude of constant delay.

The closed-loop system governed by (1)–(9) is then simulated using the following conditions

$$\begin{aligned} &M_m = M_s = 1 \text{ kg}, \ K_h = 20 \text{ N/m}, \ K_e = 1000 \text{ N/m}, \\ &T = 0.2 \text{ s}, \ B_m = B_{s1} = 0.5 \text{ Ns/m}, C_m = C_s = 0 \text{ Ns/m} \\ &b = 0.06, \ B_{s2} = 1/4b^2, \ K_s = 100, \\ &x_{md} = 0.5 \text{ m}, \ F_h^\star(t) \equiv 0 \text{ N}. \end{aligned} \tag{10}$$

The simulation result shown in Fig. 2 clearly demonstrates that positions and forces of both local and remote robots are blowing up. The reason for this unstable behavior is explained as follows. Substituting (5) and (6) into (4) and assuming that $x_{sd}(0) = x_s(0)$ results in

$$M\ddot{x}_s + C\dot{x}_s + Kx_s = K_s x_{sd} + B_{s2}\dot{x}_{sd}, \tag{11}$$

where $M = M_s$, $C = B_{s1} + B_{s2} + C_s$, and $K = K_s + K_e$. The expression (11) describes the dynamics of remote side (including the controlled remote robot and the environment) whose input is $\dot{x}_{sd}(t)$ and output is $F_e(t)$ (see also Fig. 1). Let s be the complex variable, and denote $\dot{\mathcal{X}}_{sd}(s)$ and $\mathcal{F}_e(s)$ as Laplace transforms of $\dot{x}_{sd}(t)$ and $F_e(t)$, respectively. From (11), the transfer function of the remote side is

$$G(s) = \frac{\mathcal{F}_e(s)}{\dot{\mathcal{X}}_{sd}(s)} = \frac{K_e\left(K_s + B_{s2}s\right)}{s\left(Ms^2 + Cs + K\right)}.$$

It can be seen that $G(s)$ has a relative degree of two, so $G(s)$ is non-passive [12, p. 238]. Providing that both local side

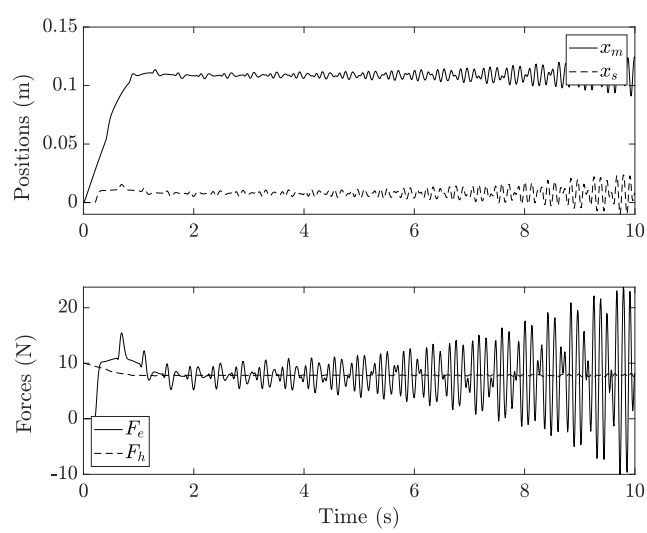

Fig. 2. Unstable WT-based teleoperation system when the contact force is directly transmitted back rather than the coordinating force.

(including the controlled local robot and the human operator) and the WT-based communication law (9) are lossless passive, the non-passivity of $G(s)$ results in the passivity shortage for the overall closed-loop system. In the energy point of view, when directly transmitting the contact force F_e back to the local side, the remote side might indefinitely inject energy into the whole system despite the fact that the environment is passive itself. This explains the aforementioned instability behavior. Note that, when the coordinating force F_s is fed back instead, the remote side is passive, leading to an overall stable system. Nevertheless, as mentioned before, sending the coordinating force F_s decreases the level of telepresence.

In fact, the transfer function $G(s)$ describing the remote side has a shortage of passivity in input (see [13]). To determine the level of this shortage, frequency domain techniques can be used [14]. However, these methods do not provide useful information about the system states that is crucial for the subsequent stability analysis. Therefore, the Lyapunov technique is employed to measure the level of passivity shortage in input of $G(s)$. Choosing the following quadratic Lyapunov function

$$V_r = \boldsymbol{x}^\top \boldsymbol{P} \boldsymbol{x}, \tag{12}$$

where $\boldsymbol{x} = [\dot{x}_s \;\; x_s \;\; x_{sd}]^\top$ and $\boldsymbol{P} > 0$ is to-be-determined. Let $\boldsymbol{z} = [\dot{x}_s \;\; x_s \;\; \dot{x}_{sd} \;\; x_{sd}]^\top$. Differentiating V_r once, then adding and subtracting the term $K_e x_s \dot{x}_{sd} + \alpha \dot{x}_{sd}^2 - \lambda \dot{x}_s^2$, yields

$$\begin{aligned} \dot{V}_r = \boldsymbol{z}^\top \left(\boldsymbol{A}_2^\top \boldsymbol{P} \boldsymbol{A}_1 + \boldsymbol{A}_1^\top \boldsymbol{P} \boldsymbol{A}_2 + \boldsymbol{A}_3 \right) \boldsymbol{z} \\ + \left(F_e \dot{x}_{sd} + \alpha \dot{x}_{sd}^2 - \lambda \dot{x}_s^2 \right). \end{aligned} \tag{13}$$

The matrices $\boldsymbol{A}_1$, $\boldsymbol{A}_2$ and $\boldsymbol{A}_3$ in (13) are given by

$$\boldsymbol{A}_1 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \; \boldsymbol{A}_2 = \begin{bmatrix} \frac{-C}{M} & \frac{-K}{M} & \frac{B_{s2}}{M} & \frac{K_s}{M} \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix},$$

$$\boldsymbol{A}_3 = \begin{bmatrix} \lambda & 0 & 0 & 0 \\ \star & 0 & -K_e/2 & 0 \\ \star & \star & -\alpha & 0 \\ \star & \star & \star & 0 \end{bmatrix}.$$

Note that, $\lambda > 0$ can be chosen to be arbitrarily small. The constant α is the level of passivity shortage in input of the remote side, and it can be determined by solving the following optimization program with linear matrix inequality (LMI) constraints

$$\begin{aligned} \min_{\alpha, \boldsymbol{P}} \quad & \alpha \\ \text{s.t.} \quad & \boldsymbol{P} > 0 \\ & \boldsymbol{A}_2^\top \boldsymbol{P} \boldsymbol{A}_1 + \boldsymbol{A}_1^\top \boldsymbol{P} \boldsymbol{A}_2 + \boldsymbol{A}_3 \leq 0. \end{aligned} \tag{14}$$

The above optimization problem can be efficiently solved by many open-source solvers. The feasibility of (14) implies that

$$\dot{V}_r \leq F_e \dot{x}_{sd} + \alpha \dot{x}_{sd}^2 - \lambda \dot{x}_s^2. \tag{15}$$

To recover the stability for the overall system, we need to compensate for the non-passivity of the remote side. Referring to Fig. 1, in order to accomplish such a compensation, the block right next to the remote side (i.e., the communication channel) should have an excessive level of passivity that is greater than α at its output port $\dot{x}_{sd}$. Clearly, the conventional lossless WT is not able to do so. Therefore, in the subsequent section, we will design an alternative WT scheme called upper strictly passive WT that can neutralize the passivity shortage of the remote side.

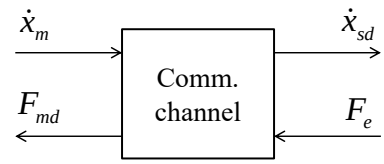

Fig. 3. The communication channel as a two-port system.

III. Upper strictly passive wave transformation design and analysis

The communication channel seen as a two-port element is shown in Fig. 3. Our goal is to design a WT-based communication channel that can provide a prescribed excessive level of passivity at the output port $\dot{x}_{sd}$ to compensate for the non-passivity of the remote side. Denote the input and output vectors of the communication channel as $\boldsymbol{u} = [\dot{x}_m \; -F_e]^\top \in \mathbb{R}^2$ and $\boldsymbol{y} = [F_{md} \;\; \dot{x}_{sd}]^\top \in \mathbb{R}^2$, respectively. Since the passivity excess is required for the upper port $\dot{x}_{sd}$, the following definition of the upper strictly passive (USP) property for two-port systems is proposed.

Definition 1 (USP)**.** A general two-port system shown in Fig. 3 is USP at a given level of $\gamma_l < 0$ and $\gamma_r < 0$ for the input–output pair $\boldsymbol{u}$–$\boldsymbol{y}$ if and only if there exists a continuously differentiable non-negative scalar function[1] $S(\boldsymbol{u}, \boldsymbol{y}, t) : \mathscr{C} \times \mathscr{C} \times \mathbb{R}^+ \mapsto \mathbb{R}^+$ such that for all $\boldsymbol{u}(t)$, $\boldsymbol{y}(t)$, and $t \geq 0$

$$\dot{S}(\boldsymbol{u}, \boldsymbol{y}, t) \leq \boldsymbol{u}^\top(t) \boldsymbol{y}(t) + \gamma_l \dot{x}_m^2(t) + \gamma_r \dot{x}_{sd}^2(t). \tag{16}$$

When $\gamma_r = 0$ is allowed, a two-port system satisfying (16) is said to be left USP (LUSP). In the same manner, if $\gamma_l = 0$ is permitted, a two-port system fulfilling (16) is said to be right USP (RUSP).

Note that, a definition for lower strictly passive property is not necessary because the two-port system can be flipped to suit the application at hand.

In contrast to a generic two-port system, the communication channel must meet another crucial property, namely the steady-state (SS) transmission requirement.

Definition 2 (SS transmission requirement)**.** Consider a two-port system shown in Fig. 3. Assume that all input/output signals of the system including $\dot{x}_m$, $\dot{x}_{sd}$, F_{md}, and F_e are in their steady states, i.e., the following limits

$$\begin{aligned} \lim_{t \to \infty} \dot{x}_m(t) := \dot{x}_m(\infty), \; \lim_{t \to \infty} \dot{x}_{sd}(t) := \dot{x}_{sd}(\infty) \\ \lim_{t \to \infty} F_{md}(t) := F_{md}(\infty), \; \lim_{t \to \infty} F_e(t) := F_e(\infty) \end{aligned}$$

[1] $S(\boldsymbol{u}, \boldsymbol{y}, t)$ can be a function of functions, i.e., a functional.

exist and are finite. Then, the two-port system is said to satisfy the SS transmission requirement if and only if

$$\dot{x}_{sd}(\infty) = \dot{x}_m(\infty) \text{ and } F_{md}(\infty) = F_e(\infty).$$

We need to endow the communication channel with not only the USP property to render the overall system stability but also the SS transmission requirement to ensure the correctness of data transmission at least in the steady state. Unfortunately, the USP property and the SS transmission requirement are two conflicting criteria, i.e., we cannot achieve both at the same time. This claim can be verified using a counterexample: when $\dot{x}_m(t) \equiv \dot{x}_{sd}(t) \equiv 1$ and $F_{md}(t) \equiv F_s(t) \equiv 1$, the SS transmission requirement is certified, but by integrating (16), the USP property implies a result that for all $t \geq 0$

$$\begin{aligned} &S(\boldsymbol{u}, \boldsymbol{y}, 0) + \int_0^t \boldsymbol{u}^\top \boldsymbol{y} d\tau \geq -\gamma_l \int_0^t \dot{x}_m^2 d\tau - \gamma_r \int_0^t \dot{x}_{sd}^2 d\tau \\ &\Longrightarrow S(\boldsymbol{u}, \boldsymbol{y}, 0) \geq -(\gamma_l + \gamma_r)\, t, \ \forall t \geq 0. \end{aligned}$$

Clearly, the above condition is false when $\gamma_l + \gamma_r < 0$ as required by the USP (or LUSP/RUSP) specification shown in Definition 1. From the energy perspective, the USP property necessitates that the communication channel must dissipate the energy at its upper ports $\dot{x}_m$ and $\dot{x}_{sd}$ for all time, whereas the SS transmission requirement reads that the energy dissipation across the communication channel should stop in the steady state, and hence the contradiction. This conflict is similar to the one between the perfect transparency and the stability of any time-delayed teleoperation systems as pointed out in [15], namely that, the stability can only be achieved at a price of sacrificing a certain level of control performance.

By acknowledging that we cannot accomplish both USP property and SS transmission requirement for the communication channel, and that stability is an uncompromising requirement for all control systems, we seek to restrict the SS transmission requirement. Fortunately, there is an appropriate way for doing so. Recalling that, for the classical WT-based teleoperation system of [2], it has been proven in [16] that along the stability, the result $\dot{x}_m(\infty) = \dot{x}_{sd}(\infty) = 0$ must hold. As a consequence, there is only one steady state condition for a stable teleoperation system, that is to say, $\dot{x}_m(\infty) = \dot{x}_{sd}(\infty) = 0$. Thus, we propose to restrict the (full) SS transmission requirement given in Definition 2 as follows.

Definition 3 (Restricted SS transmission requirement)**.** Consider a two-port system shown in Fig. 3. Assume that all input/output signals of the system including $\dot{x}_m$, $\dot{x}_{sd}$, F_{md}, and F_e are in their steady states, *and the condition* $\dot{x}_m(\infty) = 0$ *holds*. Then, the two-port system is said to satisfy the restricted SS (RSS) transmission requirement if and only if

$$\dot{x}_{sd}(\infty) = \dot{x}_m(\infty)\ (= 0) \text{ and } F_{md}(\infty) = F_e(\infty).$$

We now proceed to design a communication law fulfilling both USP property and RSS transmission requirement. The motivation comes from a graphical interpretation of the lossless WT-based communication law (9) introduced in [3]. Since such a communication law is only lossless passive (not USP), modifications are needed. Hence, a generalized WT shown in Fig. 4 is put forward, where the unknown coefficients c_{ij}

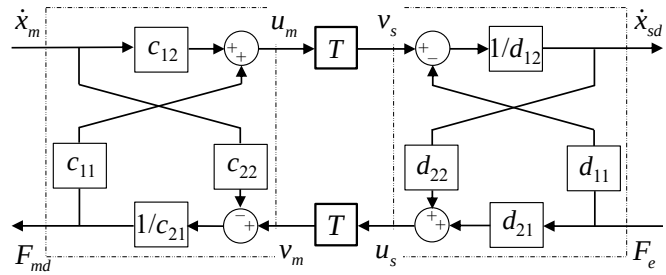

Fig. 4. The communication channel is governed by a generalized WT, where $T > 0$ is the magnitude of constant time delays.

and d_{ij} $(i, j = 1, 2)$ are to-be-determined. We firstly derive constraints for free variables c_{ij} and d_{ij} $(i, j = 1, 2)$ to realize the USP property. Afterward, additional constraints will be determined to achieve the RSS transmission requirement.

Theorem 1. By using the following storage function

$$V_c = \int_{t-T}^{t} \left[u_m^2(\tau) + u_s^2(\tau)\right] d\tau,$$

the generalized WT shown in Fig. 4 can achieve the USP property defined in Definition 1 if

$$\begin{aligned} &c_{11}^2 - c_{21}^2 = 0,\ c_{12}^2 - c_{22}^2 = \gamma_l,\ c_{11}c_{12} - c_{21}c_{22} = 0.5 \\ &d_{21}^2 - d_{11}^2 = 0,\ d_{22}^2 - d_{12}^2 = \gamma_r,\ d_{21}d_{22} - d_{11}d_{12} = -0.5, \end{aligned}$$

where $\gamma_l < 0$ and $\gamma_r < 0$ are desired levels of passivity excess of the communication channel.

Proof. see [11, Theorem 2]. □

Theorem 2. A necessary and sufficient condition for the generalized WT shown in Fig. 4 to achieve the RSS transmission requirement defined in Definition 3 is

$$c_{11} = d_{11},\ \ c_{21} = d_{21},\ \ c_{11}d_{22} - c_{21}d_{12} \neq 0.$$

Proof. see the Appendix. □

By aggregating all constraints given in Theorems 1–2, we can solve for the unknown variables c_{ij} and d_{ij} $(i, j = 1, 2)$ to make the generalized WT meet both USP property and RSS transmission requirement. These constraints read

$$\begin{aligned} &c_{11}^2 - c_{21}^2 = 0,\ c_{12}^2 - c_{22}^2 = \gamma_l,\ c_{11}c_{12} - c_{21}c_{22} = 0.5 \\ &d_{21}^2 - d_{11}^2 = 0,\ d_{22}^2 - d_{12}^2 = \gamma_r,\ d_{21}d_{22} - d_{11}d_{12} = -0.5 \\ &c_{11} = d_{11},\ \ c_{21} = d_{21},\ \ c_{11}d_{22} - c_{21}d_{12} \neq 0. \end{aligned}$$

Solving the above set of nonlinear constraints gives the following result ($b > 0$ is a free variable)

$$\begin{aligned} &c_{11} = b,\ c_{12} = b\gamma_l + \frac{1}{4b},\ c_{21} = b,\ c_{22} = b\gamma_l - \frac{1}{4b} \\ &d_{11} = b,\ d_{12} = -b\gamma_r + \frac{1}{4b},\ d_{21} = b,\ d_{22} = -b\gamma_r - \frac{1}{4b}. \end{aligned} \tag{17}$$

The generalized WT shown in Fig. 4 whose coefficients satisfying (17) is called USP wave transformation. The USP communication law is therefore given by

$$\begin{aligned}
\dot{x}_{sd}(t) &= \frac{4b^2\gamma_l + 1}{-4b^2\gamma_r + 1}\dot{x}_m(t-T) + \zeta_1 \\
F_{md}(t) &= F_e(t-T) + \zeta_2 \\
\zeta_1 &= \frac{4b^2}{-4b^2\gamma_r + 1}\left[F_{md}(t-T) - F_e(t)\right] \\
\zeta_2 &= \frac{1}{b}\left[-\left(b\gamma_l - \frac{1}{4b}\right)\dot{x}_m(t) - \left(b\gamma_r + \frac{1}{4b}\right)\dot{x}_{sd}(t-T)\right]
\end{aligned} \tag{18}$$

If we choose $\gamma_r < -\alpha$, i.e., the level of excessive passivity of the USP communication law is greater than the level of passivity shortage of the remote side, the USP communication law is able to stabilize the non-passive remote side while covering the RSS transmission requirement.

The following Theorem outlines the stability of the USP communication law (18).

Theorem 3. Assume that $\dot{x}_{sd}(t)$ is bounded for all $t < 0$. The USP communication law (18) has the following properties

1) It is bounded input–bounded output stable, namely that, when the inputs $\dot{x}_m$ and F_e belong to $\mathcal{L}_\infty$, then the outputs $\dot{x}_{sd}$ and F_{md} also belong to $\mathcal{L}_\infty$.
2) As $t \to \infty$, if $\dot{x}_m(t) \to 0$ and $F_e(t) - F_e(t-2T) \to 0$, then $\dot{x}_{sd}(t) \to 0$ and $F_{md}(t) - F_e(t-T) \to 0$.

Proof. From (18), it can be obtained that

$$\dot{x}_{sd}(t) = \delta\dot{x}_{sd}(t-2T) + \Delta(t). \tag{19}$$

The terms δ and $\Delta(t)$ in (19) are given by

$$\begin{aligned}
\delta &= \frac{-4b^2\gamma_r - 1}{-4b^2\gamma_r + 1}, \text{ (so } |\delta| < 1), \\
\Delta &= \frac{4b}{-4b^2\gamma_r + 1}\left[\frac{1}{2b}\dot{x}_m(t-T) + b\left(F_e(t-2T) - F_e(t)\right)\right].
\end{aligned}$$

The proof is trivial for $\delta = 0$. When $\delta \neq 0$, the step method [17, p. 13] can be used to solve the linear algebraic delayed equation (19). By induction, it is possible to prove that for all $m, n \in \mathbb{N}$ and for all $t \in [2qT,\, 2(q+1)T]$ where $q := m+n$, the following result holds

$$\dot{x}_{sd}(t) = \delta^{m+1}\dot{x}_{sd}(\mu) + \sum_{i=0}^{m}\delta^i\Delta(\sigma_i), \tag{20}$$

in which $\mu := t - 2(m+1)T$ and $\sigma_i := t - 2iT$. Note that, $\mu \in [2(n-1)T,\, 2nT]$ and $\sigma_i \in [2(q-i)T,\, 2(q-i+1)T]$.

We firstly prove 1). By using the assumption that the inputs $\dot{x}_m \in \mathcal{L}_\infty$ and $F_e \in \mathcal{L}_\infty$, there exists $\Delta_{\max} > 0$ such that for all $t \geq 0$, $|\Delta(t)| < \Delta_{\max}$. Choose $n = 0$. For each $t \geq 0$, there always exists $m \in \mathbb{N}$ such that $t \in [2mT,\, 2(m+1)T]$. Therefore, by (20), it is true that

$$|\dot{x}_{sd}(t)| < |\delta|^{m+1}\phi_{max} + \left(\sum_{i=0}^{m}|\delta|^i\right)\Delta_{\max}, \tag{21}$$

where $\phi_{max} := \max_{\mu\in[-2T,\,0]}|\dot{x}_{sd}(\mu)| > 0$ is bounded by assumption. Since $|\delta| < 1$, the use of geometric progression leads to the fact that, for all $m \in \mathbb{N}$, we have

$$\sum_{i=0}^{m}|\delta|^i = \frac{1-|\delta|^{m+1}}{1-|\delta|} < \frac{1}{1-|\delta|}. \tag{22}$$

Applying the result above on (21) yields

$$|\dot{x}_{sd}(t)| < k_{\max} < \infty, \tag{23}$$

where $k_{\max} := \phi_{max} + \Delta_{\max}/(1-|\delta|)$. Thus, $|\dot{x}_{sd}(t)|$ is bounded by a constant for all $t \geq 0$, namely that $\dot{x}_{sd} \in \mathcal{L}_\infty$. Since $\dot{x}_m \in \mathcal{L}_\infty$, $F_e \in \mathcal{L}_\infty$ (by assumption) and $\dot{x}_{sd} \in \mathcal{L}_\infty$, using the second equation of (18) results in $F_{md} \in \mathcal{L}_\infty$. Therefore, the claim 1) is proven.

The claim 2) will be demonstrated next. By assumption, $\dot{x}_m(t) \to 0$ and $F_e(t) - F_e(t-2T) \to 0$ as $t \to \infty$, so $\lim_{t\to\infty}\Delta(t) = 0$. By using this result and the definition of limits at infinity [18, p. 121], let $\varepsilon > 0$, there exists $\mu_t \geq 0$ such that for all $t \geq \mu_t$, $|\Delta(t)| < 0.5\varepsilon(1-|\delta|)$. Choose m and n ($m, n \in \mathbb{N}$) such that $m > (\ln(\varepsilon/2k_{\max})/\ln|\delta|) - 1$ and $n > \mu_t/(2T)$. We also define $\mu_t' := 2qT$. From (20), for all $t \geq \mu_t'$, the following result holds

$$|\dot{x}_{sd}(t)| < |\delta|^{m+1}k_{\max} + \left(\sum_{i=0}^{\infty}|\delta|^i\right)\Delta'_{\max}, \tag{24}$$

where $\Delta'_{\max} = \sup_{\omega\in[2nT,\,\infty)}|\Delta(\omega)|$. Since $\mu_t < 2nT$, it is true that $\Delta'_{\max} < 0.5\varepsilon(1-|\delta|)$. Using the chosen values of m, n, and also (22), we have $|\dot{x}_{sd}(t)| < \varepsilon$. Therefore, it was proven that, for each $\varepsilon > 0$, there exists $\mu_t' \geq 0$ such that for all $t \geq \mu_t'$, $|\dot{x}_{sd}(t)| < \varepsilon$. Thus, $\lim_{t\to\infty}\dot{x}_{sd}(t) = 0$ by definition. Using this result and also taking a limit as $t \to \infty$ for both sides of the second equation in (18) leads to the fact that $\lim_{t\to\infty}[F_{md}(t) - F_e(t-T)] = 0$. This therefore completes the proof. □

IV. Stability and performance analyses

A. Stability analysis

Theorem 4 outlines the stability result for the closed-loop system shown Fig. 1.

Theorem 4. Consider the teleoperation system shown in Fig. 1 where the local robot's dynamics (1) is controlled by (7), the remote robot's dynamics (4) is controlled by (6), and the communication channel is operated by the USP communication law (18). Then, all signals of the closed-loop system are bounded. Moreover, $\lim_{t\to\infty}\dot{x}_m(t) = \lim_{t\to\infty}\dot{x}_s(t) = 0$.

Proof. Consider the below Lyapunov–Krasovskii functional

$$\begin{aligned}
V =& \frac{1}{2}M_m\dot{x}_m^2 + \frac{1}{2}K_h\left(x_{md} - x_m\right)^2 + V_r + V_c \\
\Longrightarrow V =& \frac{1}{2}M_m\dot{x}_m^2 + \frac{1}{2}K_h\left(x_{md} - x_m\right)^2 + \boldsymbol{x}^\top\boldsymbol{P}\boldsymbol{x} \\
&+ \int_{t-T}^{t}\left[u_m^2(\tau) + u_s^2(\tau)\right]d\tau.
\end{aligned}$$

By using (1), (15) and the result of Theorem 1, the first derivative of V is given by

$$\begin{aligned}\dot{V} =& - B_m\dot{x}_m^2 - F_{md}\dot{x}_m + F_h^\star\dot{x}_m + K_h\dot{x}_m\left(x_{md} - x_m\right)\\ &- K_h\dot{x}_m\left(x_{md} - x_m\right) + F_e\dot{x}_{sd} + \alpha\dot{x}_{sd}^2 - \lambda\dot{x}_s^2\\ &+ F_{md}\dot{x}_m - F_e\dot{x}_{sd} + \gamma_l\dot{x}_m^2 + \gamma_r\dot{x}_{sd}^2.\end{aligned}$$

Simplifying the above expression results in

$$\dot{V} = -\left(B_m - \gamma_l\right)\dot{x}_m^2 + \left(\gamma_r + \alpha\right)\dot{x}_{sd}^2 - \lambda\dot{x}_s^2 + F_h^\star\dot{x}_m. \quad (25)$$

It is straightforward to verify the following inequality

$$\begin{aligned}&-\left(B_m - \gamma_l\right)\dot{x}_m^2 + F_h^\star\dot{x}_m\\ &\quad\le -\frac{1}{2}\left(B_m - \gamma_l\right)\dot{x}_m^2 + \frac{1}{2\left(B_m - \gamma_l\right)}F_h^{\star 2}.\end{aligned} \quad (26)$$

Substituting (26) into (25) yields

$$\begin{aligned}\dot{V} \le& -\frac{1}{2}\left(B_m - \gamma_l\right)\dot{x}_m^2 + \left(\gamma_r + \alpha\right)\dot{x}_{sd}^2 - \lambda\dot{x}_s^2\\ &+ \frac{1}{2\left(B_m - \gamma_l\right)}F_h^{\star 2}.\end{aligned} \quad (27)$$

As mentioned previously, we choose $\gamma_r < -\alpha$ to compensate for the non-passivity of the remote side. Integrating (27) gives

$$\begin{aligned}V(t) - V(0) \le& -\frac{\left(B_m - \gamma_l\right)}{2}\int_0^t \dot{x}_m^2 d\tau + \left(\gamma_r + \alpha\right)\int_0^t \dot{x}_{sd}^2 d\tau\\ &- \lambda\int_0^t \dot{x}_s^2 d\tau + \frac{1}{2\left(B_m - \gamma_l\right)}\int_0^t F_h^{\star 2} d\tau.\end{aligned} \quad (28)$$

Since $V(t) \ge 0$, $\forall t \ge 0$, from (28), we have

$$\begin{aligned}&\frac{\left(B_m - \gamma_l\right)}{2}\int_0^t \dot{x}_m^2 d\tau - \left(\gamma_r + \alpha\right)\int_0^t \dot{x}_{sd}^2 d\tau + \lambda\int_0^t \dot{x}_s^2 d\tau\\ &\le \frac{1}{2\left(B_m - \gamma_l\right)}\int_0^t F_h^{\star 2} d\tau + V(0).\end{aligned} \quad (29)$$

Note that $\int_0^\infty F_h^{\star 2} d\tau < \infty$ (i.e., $F_h^\star \in \mathcal{L}_2$) by an assumption made in section II and $V(0)$ is bounded due to assumed bounded initial conditions. Thus, by taking the limit as $t \to \infty$ for both sides of (29), the below integrals exist and are finite

$$\int_0^\infty \dot{x}_m^2 d\tau,\ \int_0^\infty \dot{x}_{sd}^2 d\tau,\ \int_0^\infty \dot{x}_s^2 d\tau.$$

As a consequence, $\dot{x}_m \in \mathcal{L}_2$, $\dot{x}_{sd} \in \mathcal{L}_2$, and $\dot{x}_s \in \mathcal{L}_2$. Moreover, using these results on (28) leads to the fact that $V(t) \in \mathcal{L}_\infty$, therefore

$$x_{md} - x_m,\ x_s,\ x_{sd},\ \dot{x}_m,\ \text{and}\ \dot{x}_s \in \mathcal{L}_\infty.$$

Since $F_h = K_h(x_{md} - x_m) + F_h^\star$ and $F_e = K_e x_s$ where x_{md} and $F_h^\star$ are also assumed to be bounded (see section II), the functions F_h, F_e, and x_m also belong to $\mathcal{L}_\infty$. Using Theorem 3, the results $\dot{x}_m \in \mathcal{L}_\infty$ and $F_e \in \mathcal{L}_\infty$ imply that $\dot{x}_{sd} \in \mathcal{L}_\infty$ and $F_{md} \in \mathcal{L}_\infty$. It is therefore concluded that the control laws $\tau_m \in \mathcal{L}_\infty$ and $\tau_s \in \mathcal{L}_\infty$ (see (6) and (7)). By employing all of these information, it is true that accelerations of both local and remote robots must be bounded, i.e., $\ddot{x}_m \in \mathcal{L}_\infty$ and $\ddot{x}_s \in \mathcal{L}_\infty$ (see (1) and (4)). Since $\dot{x}_i \in \mathcal{L}_2$ and $\ddot{x}_i \in \mathcal{L}_\infty$ $(i = m, s)$, it can be inferred that $\dot{x}_i(t) \to 0$ as $t \to \infty$ [19]. This completes the proof. □

We thus proved that the established USP communication law can stabilize the non-passive remote side, allowing direct contact force feedback to be used in the WT-based teleoperation architecture.

B. Performance analysis

The next Theorem demonstrates the performance of the proposed teleoperation system in the free-motion and hard-contact scenarios. Denote $e_p(t) = x_s(t) - x_m(t - T)$ and $e_f(t) = F_h(t) - F_e(t - T)$ as the position and force tracking errors, respectively.

Theorem 5. Consider the closed-loop teleoperation system as described in Theorem 4, the following results hold

1) In the free motion (i.e., when $F_e \equiv 0$), if $\gamma_r = -1/4b^2$ is selected, then $\lim_{t\to\infty} e_p(t) = x_s(0) - x_m(-T)$. Moreover, if initial positions of the two robots are identical (i.e., $x_s(0) = x_m(-T)$), the position tracking error $e_p(t) \to 0$ as $t \to \infty$.
2) In the hard-contact scenario (i.e., when $\lim_{t\to\infty} \ddot{x}_m(t) = 0$), the force tracking error $e_f(t) \to 0$ as $t \to \infty$.

Proof. First, we will demonstrate 1). Substituting the assumption $\gamma_r = -1/4b^2$ and $F_e \equiv 0$ into (19), we have $\dot{x}_{sd}(t) = \dot{x}_m(t - T)$. Therefore, for all $t \ge 0$

$$x_{sd}(t) - x_m(t - T) = x_{sd}(0) - x_m(-T). \quad (30)$$

By differentiating (4) once and noting that $\ddot{x}_m$, $\ddot{x}_s$, $\dot{x}_{sd}$, and $\dot{x}_s \in \mathcal{L}_\infty$ (see Theorem 4), the jerk $\dddot{x}_s$ of the remote robot must also belong to $\mathcal{L}_\infty$. Therefore, $\ddot{x}_s$ is uniformly continuous. Moreover, since $\lim_{t\to\infty} \dot{x}_s(t) = 0$ (see Theorem 4), it can be inferred that $\lim_{t\to\infty} \ddot{x}_s(t) = 0$ as a consequence of Barbalat lemma [12, p. 323]. Taking a limit as $t \to \infty$ for both sides of (4) yields

$$\lim_{t\to\infty}\left[x_{sd}(t) - x_s(t)\right] = x_{sd}(0) - x_s(0). \quad (31)$$

Thus, the result $\lim_{t\to\infty} e_p(t) = x_s(0) - x_m(-T)$ can be deduced from (30)–(31). Moreover, it is obvious that when $x_s(0) = x_m(-T)$, $e_p(t) \to 0$ as $t \to \infty$.

Next, the statement 2) will be proven. Taking a limit as $t \to \infty$ for both sides of (1), and noting that $\lim_{t\to\infty} \dot{x}_m(t) = 0$ (see Theorem 4) and $\lim_{t\to\infty} \ddot{x}_m(t) = 0$ (by assumption), the following result holds

$$\lim_{t\to\infty}\left[F_h(t) - F_{md}(t)\right] = 0. \quad (32)$$

Since $\dot{x}_s \in \mathcal{L}_\infty$ and $\dot{x}_s(t) \to 0$ as $t \to \infty$, the Lebesgue Dominated Convergence Theorem [18, p. 318] can be applied to the following operation

$$\begin{aligned}&\lim_{t\to\infty}\left[F_e(t) - F_e(t - 2T)\right] = K_e \lim_{t\to\infty}\int_{t-2T}^t \dot{x}_s(\tau)d\tau\\ &= K_e\int_{-2T}^0 \lim_{t\to\infty}\dot{x}_s(t + \tau)d\tau = 0.\end{aligned} \quad (33)$$

Based on the result of (33) and note that $\lim_{t\to\infty} \dot{x}_m(t) = 0$, from Theorem 3, the following result can be obtained

$$\lim_{t\to\infty}\left[F_{md}(t) - F_e(t - T)\right] = 0. \quad (34)$$

From (32) and (34), the result $e_f(t) \to 0$ as $t \to \infty$ holds. This therefore completes the proof. □

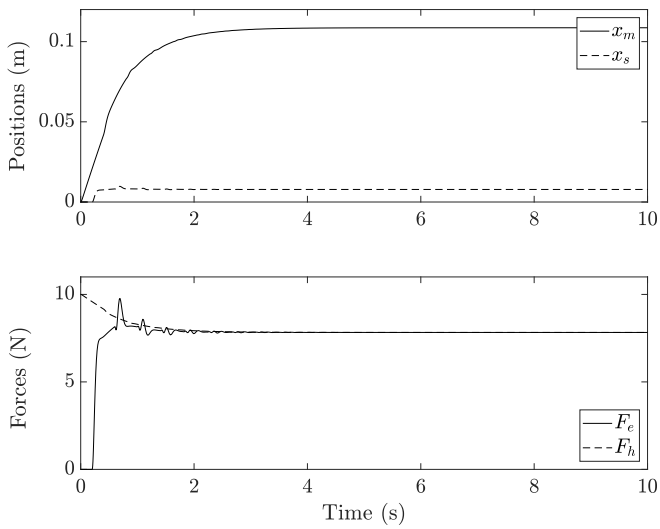

Fig. 5. Simulation result of the WT-based teleoperation system with the USP communication law and direct contact force feedback (simulation 1).

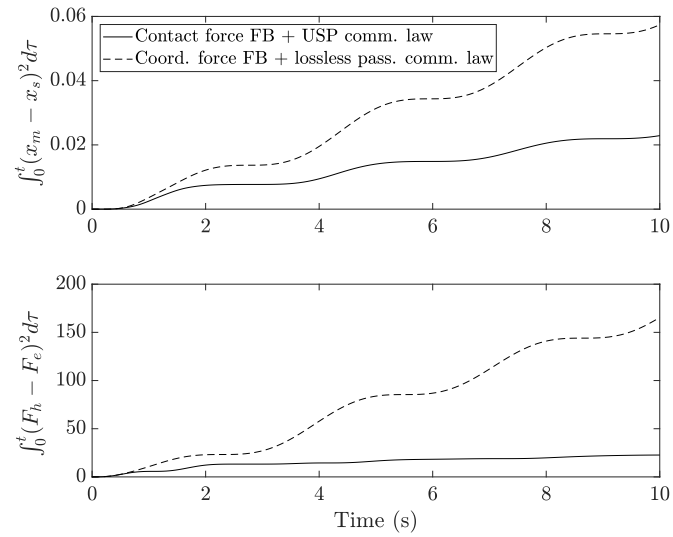

Fig. 6. Result of simulation 2.

V. Simulation results

A. Simulation 1

In this simulation, we wish to verify the ability of the USP communication law in stabilizing the WT-based teleoperation system employing direct contact force feedback. Therefore, a similar simulation as described in section II is repeated here. By employing the system parameters and control gains shown in (10) and setting $\lambda = 0.01$, the semi-definite optimization program (14) is solved by the MOSEK solver. This solver can be accessed by MATLAB using the YALMIP parser. The optimization result gives $\alpha = 9.07$. Thus, it is chosen that $\gamma_r = -20$ to realize the requirement $\gamma_r < -\alpha$ and $\gamma_l = -10$ is also selected. The simulation result is shown in Fig. 5. It can be observed that, unlike the unstable teleoperation system as found in section II when a lossless passive communication law is used, the USP communication law succesfully stabilizes the WT-based teleoperation system employing direct contact force feedback. The applied human force F_h and the contact force F_e converge to one another, meaning that the human operator can feel an accurate contact force.

B. Simulation 2

By this simulation, we wish to show the advantage of the WT-based teleoperation system using the direct contact force feedback and the USP communication law over that employing the coordinating force feedback and the lossless passive communication law. The parameters of local and remote robots used in this simulation are

$$M_m = M_s = 1 \text{ kg},\ K_h = 20 \text{ N/m},\ K_e = 10 \text{ N/m},\ T = 0.2 \text{ s}$$
$$B_m = B_{s1} = 0.5 \text{ Ns/m},\ C_m = 0 \text{ Ns/m},\ C_s = 30 \text{ Ns/m},$$
$$x_{md} = 0 \text{ m},\ F_h^\star(t) = \begin{cases} 0.5\sin(t) \text{ N}, & 0 \le t \le 10 \\ 0, & t > 10. \end{cases}$$

The conditions $C_s = 30$ Ns/m and $K_e = 10$ N/m mean that the remote robot is highly frictional and it is being in contact with a soft object. Intuitively, in such a circumstance, the coordinating force F_s is much larger than the true contact force F_e. This results in inaccurate force information feeding back to the human operator when we transmit the coordinating force rather than the contact force. To draw a fair comparison, all control gains of the two scenarios are optimized based on the following cost function

$$J = \omega_1 \int_0^{10} (x_m - x_s)^2 \, d\tau + \omega_2 \int_0^{10} (F_h - F_e)^2 \, d\tau,$$

where $\omega_1 = 10^3$ and $\omega_2 = 1$. The particle swarm optimization mechanism is employed to perform the gain optimizations. Optimal results for the case of coordinating force feedback + lossless communication law are

$$b = 0.1411,\ K_s = 497.31,\ B_{s2} = 12.5613.$$

For the case of direct contact force feedback + USP communication law, it is selected that $\lambda = 0.01$ and $\gamma_l = 0$, and the optimized control gains are

$$b = 0.1506,\ K_s = 502.76,\ B_{s2} = 11.025,\ \gamma_r = -2.1412.$$

The simulation result shown in Fig. 6 clearly demonstrates a significant advantage when employing the direct contact force feedback in tandem with the USP communication law. This therefore confirms the main result of the paper.

VI. Discussions

In order to emphasize the fundamental idea of the proposed USP communication law, simple linear models of local and remote robots are employed (see (1) and (4)). We can indeed extend our analysis to encompass the case of nonlinear local and remote robots by using some techniques. For instance, a combination of feedback linearization and model decoupling [20] can transform nonlinear robots into linear systems like (1) and (4). However, such an extension would necessitate more in-depth analysis, which is beyond the scope of this paper.

A passive (spring) environment (i.e., when $K_e \geq 0$) is considered in this paper. In fact, the proposed stabilizing communication law can handle some non-passive environments, such as those with negative damping and/or non-passive constant force. Nevertheless, it remains unclear how to cope with a negative spring environment (i.e., when $K_e < 0$). This open problem will be tackled in the future.

VII. Conclusion and Future Work

This brief establishes a new wave transformation scheme to stabilize a non-passive remote system, which is a result of employing direct environmental contact force for feedback rather than the coordinating force. The communication law based on the proposed wave transformation technique is capable of compensating for the passivity shortage of the remote side while maintaining the restricted steady-state transmission requirement. The stability of a wave transformation-based teleoperation architecture using the established stabilizing communication law is proven. It has also been demonstrated by simulations that the use of direct contact force feedback in tandem with the proposed communication law provides better experience of telepresence to the human operator. Our future work will focus on further applications of the proposed wave transformation technique in position–force bilateral teleoperations with non-passive environment models as well as multilateral telecontrols.

Appendix
Proof of Theorem 2

Since the RSS transmission requirement is being considered, it should make clear that the following conditions are presumed throughout the proof

- All signals $\dot{x}_m$, $\dot{x}_{sd}$, F_{md}, and F_e are in steady state.
- $\dot{x}_m(\infty) = 0$.

From the left generalized WT of Fig. 4, it can be inferred that

$$\begin{aligned} u_m(t) &= c_{11}F_{md}(t) + c_{12}\dot{x}_m(t) \\ v_m(t) &= c_{21}F_{md}(t) + c_{22}\dot{x}_m(t). \end{aligned} \tag{35}$$

Similarly, from the right generalized WT, we have

$$\begin{aligned} u_s(t) &= d_{21}F_e(t) + d_{22}\dot{x}_{sd}(t) \\ v_s(t) &= d_{11}F_e(t) + d_{12}\dot{x}_{sd}(t). \end{aligned} \tag{36}$$

The relations $v_s(t) = u_m(t-T)$ and $v_m(t) = u_s(t-T)$ also hold. Therefore, from (35)–(36), it can be obtained that

$$\begin{aligned} d_{11}F_e(\infty) + d_{12}\dot{x}_{sd}(\infty) &= c_{11}F_{md}(\infty) \\ d_{21}F_e(\infty) + d_{22}\dot{x}_{sd}(\infty) &= c_{21}F_{md}(\infty). \end{aligned} \tag{37}$$

The necessary condition can be deduced by assuming that the RSS transmission requirement is realized by the generalized WT. Thus, from Definition 3, $\dot{x}_{sd}(\infty) = \dot{x}_m(\infty) = 0$ and $F_{md}(\infty) = F_e(\infty)$ hold. Applying this result on (37) yields

$$\begin{aligned} (d_{11} - c_{11})\, F_e(\infty) &= 0 \\ (d_{21} - c_{21})\, F_e(\infty) &= 0 \end{aligned}, \quad \forall F_e(\infty) \in \mathbb{R}.$$

The equations above imply that $c_{11} = d_{11}$ and $c_{21} = d_{21}$. This is the result of necessary condition.

To determine the necessary and sufficient condition, we substitute the obtained necessary condition into (37)

$$\begin{aligned} c_{11}\left[F_e(\infty) - F_{md}(\infty)\right] + d_{12}\dot{x}_{sd}(\infty) &= 0 \\ c_{21}\left[F_e(\infty) - F_{md}(\infty)\right] + d_{22}\dot{x}_{sd}(\infty) &= 0. \end{aligned}$$

The above set of linear equations has a unique solution of $F_e(\infty) - F_{md}(\infty) = 0$ and $\dot{x}_{sd}(\infty) = 0$ if and only if $c_{11}d_{22} - c_{21}d_{12} \neq 0$. Combining this result with the necessary condition $c_{11} = d_{11}$ and $c_{21} = d_{21}$ gives the necessary and sufficient condition for the generalized WT to achieve the RSS transmission requirement. This completes the proof.

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